Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · TypeScript · React · FastAPI · AWS · LangGraph · PostgreSQL · Docker · CI/CD · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience building full-stack systems at scale - Python backend, React/TypeScript frontend, and AWS cloud infrastructure. At Entrata I owned mission-critical systems serving 100K+ users and 1M+ property records for 7 years at 99.5% uptime, shipping 20+ production modules across 120+ Agile sprints with zero compliance violations. Currently building AI-powered systems: AgentProbe (open-source AI evaluation platform - FastAPI, LangGraph, RAG pipelines, CI-integrated quality gate) and LeaseI (8-agent LangGraph workflow with RAG, HITL guardrails, 93.6% precision across 1,240 test cases). Strong in system design, API development, distributed systems, observability, and technical mentorship. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr), TypeScript, JavaScript, React.js, FastAPI, Flask, PHP, SQL, PL/SQL, R, Bash, HTML/CSS

AI & Machine Learning: LangGraph, LangChain, LLM Integration, RAG Pipelines, Vector Databases (ChromaDB), Embeddings, Semantic Search, Multi-Agent Orchestration, Tool Calling, LLM Evaluation, LLM-as-Judge, Hallucination Detection, HITL Guardrails, Prompt Engineering, Anthropic API, OpenAI API, Ollama, MCP, PyTorch, scikit-learn, XGBoost, NLP, BERT, spaCy

Backend, Data & APIs: REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Redis, NoSQL, SQLite, Performance Optimization, Query Optimization, Indexing, Idempotency, Reliability Engineering, ETL, Data Pipelines

Cloud, DevOps & Observability: AWS (S3, EC2, RDS, Glue, Lambda), Docker, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Performance Testing, Regression Testing, Observability, Structured Logging, Distributed Tracing, Monitoring, Alerting, Incident Response, Root-Cause Analysis, Agile/Scrum, Linux, Git

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer

- Designed and shipped a production AI application (LangGraph agentic pipeline, FastAPI backend, Python) as sole technical owner - built RAG retrieval layer, evaluation pipeline, CI validation gate, HITL guardrails, and observability infrastructure; delivered from zero to production in weeks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Backend systems at scale** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ users and 1M+ property records; RabbitMQ async event-driven processing with retry logic and idempotency guarantees; 200+ PostgreSQL/MySQL/Oracle query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); AWS S3/EC2/RDS/Glue infrastructure.
- **Full-stack delivery & CI/CD** - shipped 20+ production modules (Python, PHP, JavaScript, PostgreSQL) across 120+ Agile sprints; built React.js + TypeScript frontend features; GitHub Actions CI/CD with automated unit, performance, and regression testing; 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines; zero security incidents across full ownership period.
- **Observability & reliability** - built structured logging, monitoring, and alerting infrastructure; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ modules over 120+ sprints; zero compliance violations across 7yr ownership.
- **Technical leadership** - wrote design documents for 20+ production modules; mentored 5+ engineers on system design and production best practices; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (Python, PHP, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific document templates with merge fields.

PROJECTS

AgentProbe: Open-Source AI Evaluation Toolkit LangGraph · RAG · FastAPI · CI/CD · LLM Evaluation · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation platform built end-to-end: 15-endpoint FastAPI backend, React.js + TypeScript frontend, multi-provider LLM routing (Anthropic, OpenAI, Ollama), ChromaDB RAG pipeline, and 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency). LLM-as-Judge scoring with pass@k multi-trial harness; CI-integrated regression gate (GitHub Actions) blocking quality degradation before deployment; structured distributed tracing across all agent steps. Zero regressions across 1,240+ test cases. Open-source.

LeaseI: Agentic AI Support Investigation Platform LangGraph · RAG · HITL · FastAPI · MCP · Multi-Agent

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI platform with three user-facing surfaces (Document Intelligence, Support Investigation, Tenant Self-Service) sharing one SQLite knowledge repository. 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing, tool calling, and MCP integration for live regulatory data. Hand-rolled TF-IDF RAG with 100% citation correctness; deterministic HITL guardrails enforcing human oversight on regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: NLP-Driven Medical Records Query System Full Stack · AWS Lambda · HIPAA · LLM · React · PostgreSQL

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application: React.js + TypeScript frontend, Python/Flask REST API backend, LLM-powered NLP translating natural-language queries (EN/ES) into FHIR API calls using BERT and embeddings; AWS Lambda serverless backend with API gateway; HIPAA-compliant RBAC with full audit trail; PostgreSQL data services; Docker containerization; observability stack. ~60% reduction in manual clinical retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017